

Development of a Quad Rotor Tail-Sitter VTOL UAV without Control Surfaces and Experimental Verification

Atsushi Oosedo¹, Satoko Abiko¹, Atsushi Konno², Takuya Koizumi¹, Tatuya Furui¹ and Masaru Uchiyama¹

Abstract—This paper presents development of a quad rotor tail-sitter VTOL UAV (Vertical Takeoff and Landing Unmanned Aerial Vehicle) which is composed of four rotors and a fixed wing. The conventional VTOL UAVs have a drawback in the accuracy of the attitude control in stationary hovering because they were developed based on a fixed-wing aircraft and they used the control surfaces, such as aileron, elevator, and rudder for the attitude control. To overcome such a drawback, we developed a quad rotor tail-sitter VTOL UAV. The quad rotor tail-sitter VTOL UAV realizes high accuracy in the attitude control with four rotors like a quad rotor helicopter and achieves level flight like a fixed-wing airplane. The remarkable characteristic of the developed quad rotor tail-sitter VTOL UAV is that it does not use any control surfaces even in the level flight. This paper shows the design concept of the developed UAV and experimental verification of all flight modes including hovering, transition flight and level flight.

I. INTRODUCTION

Recently, UAVs are widely used to gather various types of information from the sky above for a variety of purposes in civil applications. In disasters, UAVs are expected to play very significant roles to rescue victims, explore the disaster areas or to deliver relief supplies to the isolated areas alone. Especially, small VTOL UAVs are very promising in such hazardous situations since they can fly long distance as fixed-wing airplanes and can hover as conventional rotary-wing aircraft by themselves.

Among the several types of VTOL aircraft (tilt-rotor, vector-thrust etc.) [1], a tail-sitter aircraft is the simplest way to achieve the VTOL maneuver since it does not require extra actuators for the VTOL maneuver. The tail-sitter VTOL aircraft can achieve both level flight and hovering by changing its pitch angle of the fuselage by 90° as shown in Fig. 1.

Up to now, several types of tail-sitter VTOL UAVs have been developed. Stone *et al.* developed the T-Wing tail-sitter UAV with a canard wing and tandem rotors [2]. Kita *et al.* developed a simple tail-sitter VTOL UAV with a single propeller R/C airplane [3]. The above mentioned tail-sitter VTOL UAVs were developed based on fixed-wing aircraft. Therefore, the attitude is controlled by control surfaces, such as aileron. As a result, they had relatively poor performance in stationary hovering compared to a rotary-wing aircraft, although they achieved high stability in level flight.

¹A. Oosedo, S. Abiko, T. Koizumi, T. Furui and M. Uchiyama are with Department of Mechanical Systems and Design, Graduate School of Engineering, Tohoku University, 6-6-01 Aramaki-aza-Aoba, Aoba-ku, Sendai, 980-8579, Japan. {oosedo, abiko, uchiyama}@space.mech.tohoku.ac.jp

²A. Konno is with Divi. of System Science and Informatics, Hokkaido University, 14-9, Kita-ku, Sapporo, 060-0814, Japan. {konno}@ssi.ist.hokudai.ac.jp

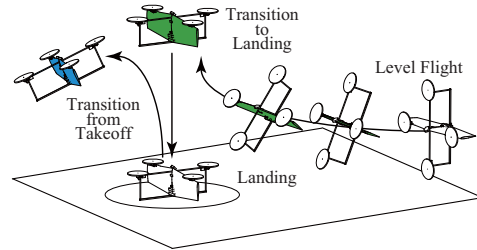


Fig. 1. Take-off and landing of a tail sitter VTOL

On the other hand, quad rotor helicopters have achieved high stability in positioning and attitude control with a simple mechanism. Therefore, several researchers have commonly used the quad rotor helicopters for their research applications, such as autonomous navigation in indoor environments and multi-vehicle flight testbed [4][5]. However, it is difficult for quad rotor helicopters to fly long distance and long duration because the most of the thrusts are consumed to lift the body up, and hence horizontal component of the thrust is small.

One solution to overcome the above drawbacks is, so called, a quad rotor tail-sitter VTOL UAV. The quad rotor tail-sitter VTOL UAV equips with a fixed wing on the basis of the quad rotor helicopter. The quad rotor tail-sitter VTOL UAV provides the ability of long distance flight while holding high stability in positioning and attitude control.

Young *et al.* developed a quad rotor tail-sitter UAV with an R/C airplane and succeeded in hovering and forward flight in 2002 [6]. However, they did not discuss autonomous level flight at all and have not reported further development since then. Sinha *et al.* developed a quad rotor tail-sitter VTOL UAV named Quadshot in 2012 [7]. Quadshot performed high dynamic maneuverability by use of a combination of differential thrust and two control surfaces, namely elevon. Hence, this UAV requires six actuators for flying. However, in principle, the quad rotor tail-sitter UAV does not require additional control surfaces since only four rotors can realize stable attitude control.

In our previous research published in [8], we performed flight simulation of a quad rotor tail-sitter UAV without using any control surfaces. The simulation result apparently showed that the UAV is able to realize level flight without control surfaces. Moreover, it is clearly shown from the simulation results that the quad rotor tail-sitter UAV can fly three times longer distance compared with that of the conventional quad rotor helicopter.

This paper presents the development of a quad-rotor tail-sitter UAV that consists of four rotors and a fixed wing based

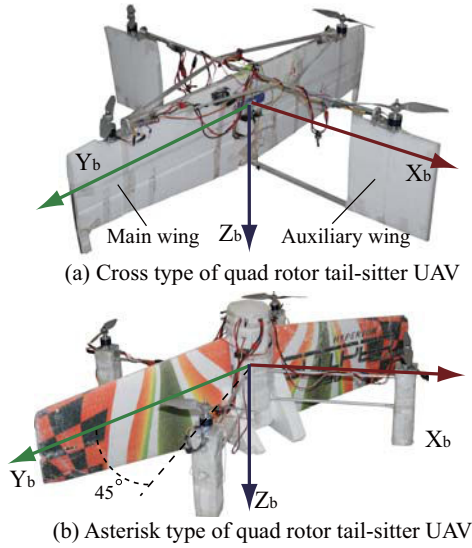


Fig. 2. Two types of the developed quad rotor tail-sitter UAVs

on the conventional quad rotor helicopter. The remarkable characteristic of the developed UAV is that it does not use any control surfaces, but it can realize all flight modes, namely hovering, transition and level flight with simple mechanism.

Firstly, we describe design concept and system configuration of the developed UAV. Secondly, we briefly describe flight control system implemented to the system, which has been developed in our previous research [9]. Finally, experimental verification of all flight modes are carried out. In the experimental verification, firstly, we compare the performance of two types of UAVs in the attitude control since the effect of slipstream was not modeled in the simulation in [8]. Then, the UAV which has better performance is selected to demonstrate the transition and level flights.

II. SYSTEM CONFIGURATION

A. Design concept of the system

Fig. 2 shows overview of the developed quad rotor tail-sitter UAVs. The quad rotor tail-sitter UAV in Fig. 2(a) is named here *cross-type* of UAV, in which the main wing is aligned under the propellers. The quad rotor tail-sitter UAV in Fig. 2(b) is named here *asterisk-type* of UAV, in which the main wing is allocated on the place rotated by 45° around the Z_b axis with respect to the allocation of the propellers.

If the control surfaces are used to control the attitude as the T-wing and Quadshot, the control surfaces must be placed under the propellers. On the other hand, if this is not the case, the main wing is placed at the area where the propeller slipstream does not affect the control performance. As one can see in Fig. 2, the cross-type of UAV has a main wing under the propellers. Therefore, one can imagine that the slipstream influences the performance of attitude control. In the simulation analysis of the quad-rotor tail-sitter VTOL UAV in [8], we did not include the detail effect of the slipstream. Therefore, we quantitatively analyze the influence of the slipstream through experiment verification

by comparing the performance of the cross-type and asterisk-type of UAVs. The detail analysis of this is shown in Section IV.

B. Coordinate Systems

In each UAV, the coordinates of the aircraft body (X_b , Y_b , Z_b) are defined as shown in Fig. 2. The rotations about the X_b axis, Y_b axis and Z_b axis are defined as roll, pitch, and yaw, respectively. The earth fixed coordinates (inertial coordinates) define X_i axis as true north, Y_i axis as east, and Z_i axis as perpendicular downward. X_b axis of the aircraft coordinates coincides with X_i axis of the inertial coordinates when pitch and yaw angles are equal to 0. The coordinate system of the aircraft is consistent in every flight modes.

C. Fuselage

The following subsections show the common specification of both developed quad rotor tail-sitter UAVs. The airframe is made of aluminum alloy and EPP (Expanded PolyProylene). The main wing is a part of a commercially available R/C airplane and the airfoil is NACA0010 whose span is 0.99 [m], the chord is 0.28 [m], and taper ratio is 0.6. Since we do not use the aileron, we fixed the aileron to prevent it from moving in this research.

A fixed pitch propeller and a brushless DC motor are used as propulsion units. The propeller diameter is 0.205 [m] and the pitch is 0.152 [m]. As a result, a thrust to weight ratio is more than or equal to 1.39.

D. Electronics

The UAV is equipped with the following processing units and sensors.

The main computer is commercially available microcomputer board (Alpha project Co., STK-7125) which has an SH2 microcomputer (Rnesas Technology Co.). This computer executes control computation, sensor information processing and transmission of a command signal (PWM:Pulse Width Modulated) to each motor. The cycle of control computation is 50 [Hz]. Flight logs are recorded on a micro-SD card.

The attitude, three-axis angular velocity and translational acceleration are measured by 3DM-GX1 produced by Microstrain Co. The sampling time is 100 [Hz]. The maximum angle error is $\pm 2^\circ$. GPS 18-5Hz from Garmin Co. is used to obtain the position of the UAV. The accuracy of horizontal direction is 4–5 meters and the accuracy of vertical position is 10–20 meters. Therefore, altitude is mainly obtained from one ultrasonic distance sensor (USS) for precise measurements. The resolution of the USS is 0.025 meter and the measurable range of the USS is 0 to 6.45 meter. One ultrasonic distance sensor and one R/C servo are mounted at the tail of the UAV. The attitude of tail-sitter UAV greatly varies in each flight mode. Therefore, in order to measure altitude with one sensor, the servo rotates the USS around the Y_b axis. The movement of this servo does not affect the attitude of the aircraft.

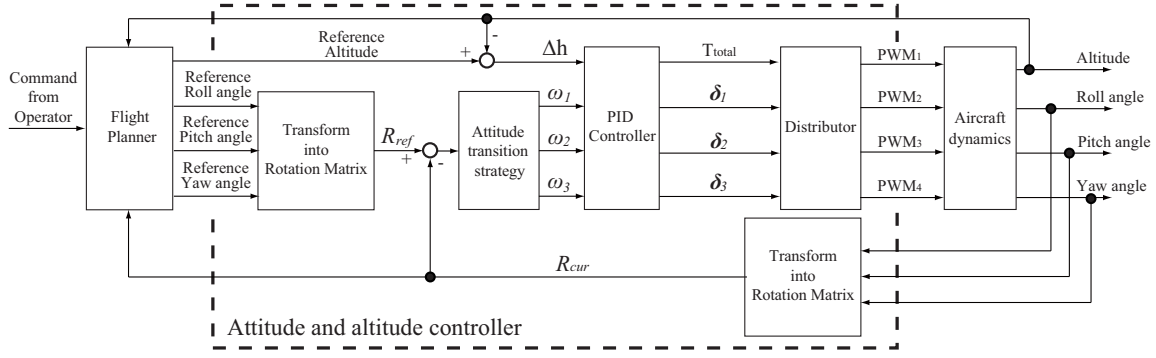


Fig. 3. Flight controller

III. CONTROL SYSTEM

A. Flight control system

Fig. 3 shows the block diagram of the flight controller. The flight control system is designed based on the PID controller.

Firstly, the flight plan and reference parameters are installed in the “Flight planner”. After the UAV receives start command from an operator, the planner generates the reference attitude and altitude based on the prearranged flight plan and sensor information including attitude and altitude. Then, these references and current attitude are transformed into rotation matrices, R_{ref} and R_{cur} , which are sent to the “Attitude transition strategy”.

In the tail-sitter aircraft, quaternion feedback control is often used due to no singular point. However, when the attitude error is large, the quaternion feedback control may fail to stabilize the UAV. Therefore, in this paper, “Resolved Tilt-Twist Angle Feedback Control”, proposed in our previous research [9], is used to calculate the attitude error for a tail-sitter VTOL UAV. The resolved tilt-twist angle feedback control increases stability against large attitude disturbance. Section III.B briefly reviews this control method.

The block of “Attitude transition strategy” generates errors around the X_b , Y_b and Z_b axis of the coordinates of the aircraft body. These errors are defined as ω_j , ($j = 1 \sim 3$). These attitude errors and altitude error are sent to the PID controller. The PID controller generates the desired differential thrusts for X_b , Y_b axis control, the desired torque for Z_b axis control and desired total thrust for altitude control. These desired difference thrusts, torque and total thrust are given as follows:

$$\delta_j = \left(K_P \omega_j + K_I \int \omega_j dt + K_D \dot{\omega}_j \right), \quad (j = 1 \sim 3) \quad (1)$$

$$T_{total} = \left(K_P \Delta h + K_I \int \Delta h dt + K_D \dot{\Delta h} \right) + mg, \quad (2)$$

where δ_1 , δ_2 , and δ_3 are the desired differential thrust for X_b axis control, Y_b axis control, and the desired torque for Z_b axis control, respectively. T_{total} is the desired total thrust and Δh denotes the error between reference and current altitude. The PID gains are provided by the ultimate sensitivity method, and empirically tuned up.

These values are sent to the distributor, which calculates revolution speed of each motor and transforms into control

command (PWM signal). The calculated PWM signals are finally sent to each motor.

B. Attitude transition strategy

This subsection briefly reviews “Resolved Tilt-Twist Angle Feedback Control” [9]. In the above control sequence, the attitude error is resolved into the tilt and twist angles. The tilt angle is composed of two angles of orthogonal axes. The method is composed of the following steps.

In the first step, the pitch and yaw errors are derived in the analogy of inverted pendulum. The pitch and yaw errors provide the tilt angle of the aircraft. Firstly, the error rotation matrix R_E between the reference orientation R_{ref} and the current orientation R_{cur} is determined as follows:

$$R_E = R_{cur}^T R_{ref} = \begin{bmatrix} R_{E11} & R_{E12} & R_{E13} \\ R_{E21} & R_{E22} & R_{E23} \\ R_{E31} & R_{E32} & R_{E33} \end{bmatrix}, \quad (3)$$

$$R_{ref} = [e_{xr} \quad e_{yr} \quad e_{zr}], \quad (4)$$

$$R_{cur} = [e_{xc} \quad e_{yc} \quad e_{zc}]. \quad (5)$$

The elements of X_b axis in R_E gives pitch and yaw errors as follows:

$$\theta_Y = \text{atan2}(R_{E31}, R_{E11}), \quad (6)$$

$$\theta_Z = \text{atan2}(R_{E21}, R_{E11}). \quad (7)$$

Then, θ_Y and θ_Z define the tilt angle θ_{tilt} as follows:

$$\theta_{tilt} = \sqrt{\theta_Y^2 + \theta_Z^2}, \quad (8)$$

In the second step, the roll error is derived. The attitude of the aircraft after rotation of θ_{tilt} becomes as follows:

$$R_v = \begin{cases} \exp(\hat{\mathbf{v}}\theta_{tilt}) & , \text{ for } R_E \neq E \\ E & , \text{ for } R_E = E \end{cases} \quad (9)$$

where E is a 3×3 identity matrix, $\mathbf{v}$ is the rotation axis vector given by normalized cross product of e_{xr} and e_{xc} as follows:

$$\mathbf{v} = \frac{e_{xc} \times e_{xr}}{|e_{xc} \times e_{xr}|} \equiv [v_x \quad v_y \quad v_z]^T, \quad (10)$$

The hat operator $\{\cdot\}$ indicates a skew-symmetric matrix.

The UAV attitude R_P after θ_{tilt} attitude change is given by using R_v as follows:

$$R_P = R_v R_c \equiv [e_{xp} \quad e_{yp} \quad e_{zp}], \quad (11)$$

where e_{jp} ($j = x, y, z$) are the unit vectors along j axis of the body coordinate frame after compensating the tilt with respect to the inertial coordinate frame. The absolute roll error is defined as follows:

$$\theta_{twist} = \cos^{-1} \left(\frac{e_{zp} \cdot e_{zr}}{|e_{zp}| |e_{zr}|} \right) \quad (12)$$

Since the range of roll angle of the aircraft is from -180° to 180° , the sign of the roll error must be identified. In order to identify the sign of the roll error θ_X , θ_{sign} is defined as follows:

$$\theta_{sign} = \cos^{-1} \left(\frac{e_{yp} \cdot e_{zr}}{|e_{yp}| |e_{zr}|} \right) \quad (13)$$

By using θ_{sign} , the roll error θ_X of the UAV is identified as follows:

$$\theta_X = \begin{cases} \theta_{twist} & , \text{ for } \theta_{sign} \leq \frac{\pi}{2} \\ -\theta_{twist} & , \text{ for } \theta_{sign} \geq \frac{\pi}{2} \end{cases} \quad (14)$$

In the third step, the errors around the X_b , Y_b and Z_b axis on the aircraft body coordinates are calculated to compensate each axis error simultaneously. Therefore, the pitch and yaw errors in the inertial coordinates must be projected onto the aircraft body coordinates, which is expressed by rolling θ_X around X_b axis in the inertial coordinates as follows:

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_X & -\sin \theta_X \\ 0 & \sin \theta_X & \cos \theta_X \end{bmatrix} \begin{bmatrix} \theta_X \\ \theta_Y \\ \theta_Z \end{bmatrix} \quad (15)$$

The above errors are finally sent to the PID controller.

IV. EXPERIMENTAL VERIFICATION

This section presents experimental verification of the developed quad rotor tail-sitter VTOL UAVs. Firstly, we compare the performance of the attitude control between the cross-type of UAV and the asterisk-type of UAV to qualitatively analyze the effect of slipstream in the flight. Then, the UAV with better performance is used to demonstrate transition and level flight.

A. Effect of slipstream

In our previous research in [8], we confirmed that it is possible to develop the quad rotor tail-sitter VTOL UAV without using any control surfaces. However, the simulation did not include the influence of the slipstream in detail. Therefore, to experimentally analyze the effect of the slipstream, we developed two types of UAVs, namely cross-type and asterisk-type of UAVs as shown in Section II. As one can imagine, when the main wing is allocated right under the propellers, the slipstream generates forces to rotate the airframe to opposite direction from the rotational direction produced by the anti-torque of the propellers around the Z_b axis since the slipstream hits on the surface of the main wing. In the cross-type of UAV equipped with only main wing, the effect of the anti-torque of the propellers right above the main wing is counteracted by the effect of the slipstream. Therefore, the UAV always easily rotate in one direction. Accordingly, the effect of the slipstream leads to the difficulty of the attitude control. To solve this problem,

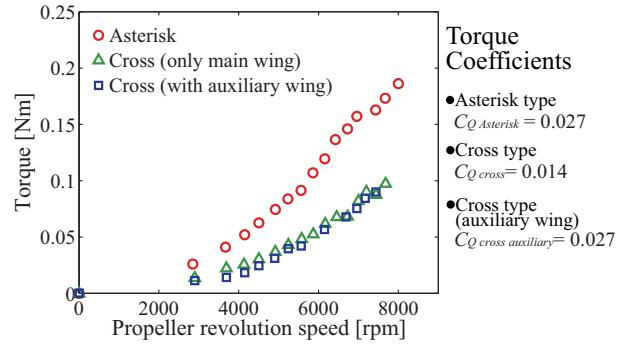


Fig. 4. Torque of slipstream and torque coefficients

there are two solutions. One is to attach auxiliary wings under the other propellers in the cross-type of UAV. Another solution is to place the main wing in the area where the slipstream does not affect to the attitude control, which is here the asterisk-type of UAV.

In the following, we investigate the effect of the slipstream by observing the change of the torque. The change of the torque around the Z_b axis is observed by a torque sensor on which the airframe is completely fixed. Fig. 4 shows the experimental result in which the torque generated around the Z_b axis is plotted with respect to the revolution speed of the propeller. Fig. 4 clearly shows that the torque of the cross-type of UAV is drastically reduced compared to that of the asterisk-type of UAV. In general, the torque generated by the propellers is expressed by the following equation.

$$Q = \rho n^2 D^5 C_Q \quad (16)$$

where n stands for the revolution speed of the propeller, ρ is the atmospheric density and D is the propeller diameter. The torque coefficient, C_Q depends on the propeller form and indicates the performance of the propeller. Fig. 4 shows the torque coefficient of two different types of UAV. As shown in Fig. 4, the asterisk-type of UAV performs almost twice better than the cross-type of UAV.

B. Attitude control capability

We also carried out experiment of attitude control in autonomous hovering mode with the cross-type of UAV, the asterisk-type of UAV and the conventional quad rotor UAV which has no wing. The total weight of the cross-type of UAV is 1.18 [kg], the asterisk-type of UAV is 1.39 [kg] and the conventional quad rotor UAV is 1.19 [kg].

In the experiment, the reference roll, pitch and yaw angles are 0° . The flight experiment was performed outdoors, and wind was blowing from the east to the west about $0.5 \sim 1.0$ [m/s]. After flying for twenty-three seconds, the experiment was terminated and these UAVs landed on the ground.

Figs. 5 to 7 show the attitude profiles in the experiment. In the figures, the dashed line depicts the reference attitude, the red dotted line depicts the attitude of the cross-type of UAV, the green solid line depicts the attitude of the asterisk-type of UAV and the blue bold line depicts the attitude of the conventional quad rotor UAV, respectively. As can be seen from Figs. 5 and 6, these UAVs follow up the reference

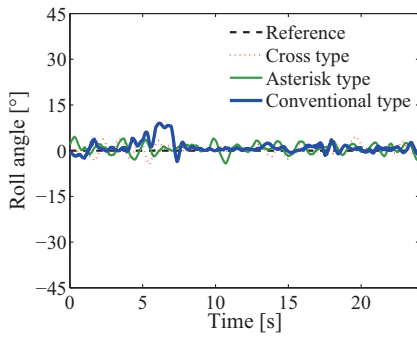


Fig. 5. Comparison of roll angle

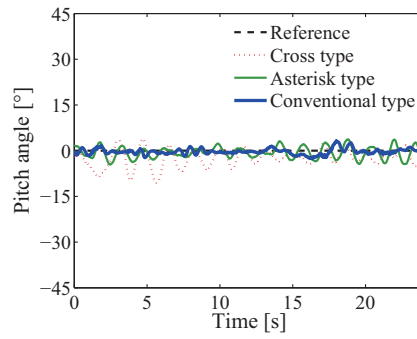


Fig. 6. Comparison of pitch angle

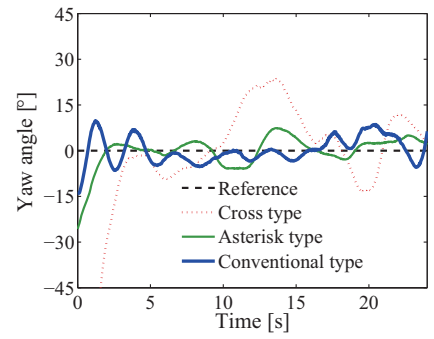


Fig. 7. Comparison of yaw angle

on the roll and the pitch. In the yaw axis, the attitude of the asterisk-type of UAV and the conventional quad rotor UAV converge to the reference while the cross-type of UAV did not converge to the reference in the experiment. The yaw control capability of both the asterisk-type of UAV and the conventional quad rotor UAV was almost same in the experiment. However, in the case when large disturbance occurs such as strong wind, it can be predicted that the yaw control capability of the asterisk-type of UAV should be degraded.

In any case, the attitude of the asterisk-type of UAV is more stable than that of the cross-type of UAV. Especially, the attitude of the asterisk-type of UAV in the yaw axis shows fast response compared to that of the cross-type of UAV. Therefore, the experiment of transition and level flight is carried out with the asterisk-type UAV.

C. Experiment of transition and level flight

The transition and level flight control strategy are experimentally verified using the asterisk-type of UAV. The block diagram of the flight controller is illustrated in Fig. 3. This flight experiment was performed outdoors, and wind was blowing from the east to the west at 1.0 [m/s].

The UAV precedes flight according to the following pre-defined flight plan.

Step.1–Lifting up to start experiment: The UAV is lifted up to about 6 [m] in hovering with manual control mode.

Step.2–Switching control mode: The control mode is automatically switched to autonomous control.

Step.3–Stationary hovering: The UAV performs autonomous stationary hovering on site for 2 seconds. At this time, the reference roll and pitch are 0 [°] and the reference yaw is -35 [°].

Step.4–Transition to level flight: After stationary hovering, the UAV initiates transition flight to the level flight. At this time, the reference roll is 0 [°], yaw is -35 [°] and pitch angle is -75 [°]. Step.4 continues until the pitch angle reaches -75 [°].

Step.5–Level flight: After the pitch angle reaches -75 [°], the UAV flies with the above mentioned reference attitude and altitude for 8 seconds. The reference pitch angle is -75 [°] and reference altitude is 6 [m].

Step.6–Transition to hovering to terminate the experiment: After the 8 second level flight, the UAV transits to hovering flight and flies down to ground.

Fig. 8 shows snapshots of the transition and level flight experiment. Please refer to the attached movie for more information. The UAV begins to lift up in manual control mode at $t = 0$ [s]. At 20.2 seconds after the takeoff, the control mode switches to autonomous control. As can be seen in Fig. 8, the UAV maintained almost reference attitude, but the altitude significantly changed after transition from the hovering to the level flight. This cause is discussed later. After flying for eight seconds in level flight mode, the UAV switched to hovering to terminate the experiment and the UAV landed on the ground. The total flight time was thirty three seconds.

Figs. 9 to 11 show the attitude profiles during the experiment. The attitude on every axes was successfully controlled and it did not have large error until the end of the level flight. During the transition flight from the hovering to the level flight, which was started at $t = 22.2$ [s], the pitch reached the desired angle, -75 [°], at $t = 23.0$ [s]. As shown in the figure, the level flight was successfully accomplished with constant pitch angle. The maximum pitch error was 7 [°] in the level flight.

After the level flight, attitude was disturbed from $t = 30.2$ [s] due to the aerodynamics drag generated by transition flight against the fixed wing. By rapidly lifting the pitch up, unequal huge drag was generated in right and left side of the main wing.

In the experiment, the altitude of the UAV was also controlled by following the controller shown in Fig. 3. However, the altitude significantly fell in transition flight and it changed in level flight. The cause of altitude loss in transition flight is a lack of wing lift. The transition flight was performed quickly, resulting in producing very little flight speed. At $t = 23.0$ [s], the calculated flight speed is about 5.3 [m/s], and the vertical component of the lifting force (wing lift and rotor thrust) is 8.2[N]. As a result, the wing lift is insufficient to compensate for the loss in rotor lifting thrust. Hence, the UAV significantly descended. In addition, the reason of the altitude change in level flight is that the USS could not measure the proper distance from the UAV to the ground due to grass field condition. Nevertheless, we can see from Fig. 8 that the altitude in level flight was lifted up. When we analyze the thrust distribution between the vertical and horizontal component from the total thrust, the maximum vertical component of thrust becomes 4.8 [N]. This indicates that the controller generated the lift force

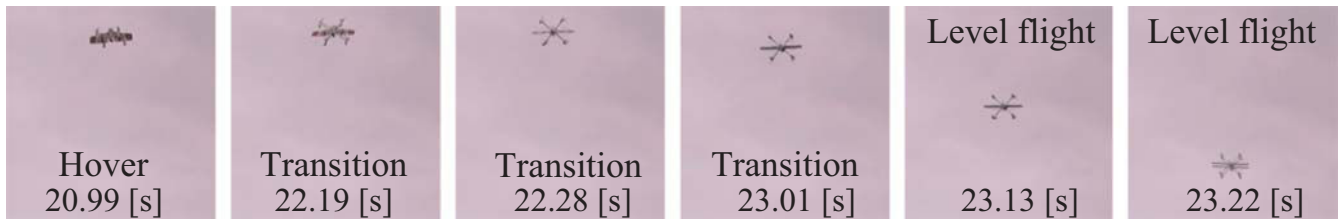


Fig. 8. Sequential photographs of the transition and level flight

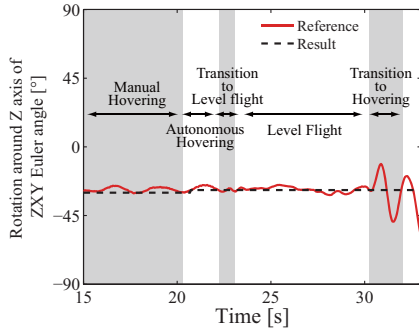


Fig. 9. Rotation around Z axis of ZXY Euler angle

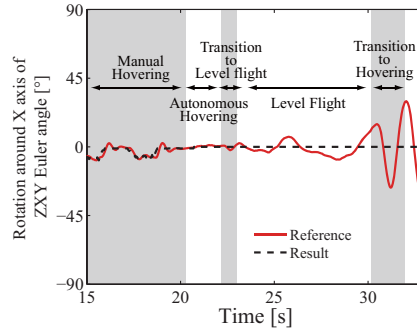


Fig. 10. Rotation around X axis of ZXY Euler angle

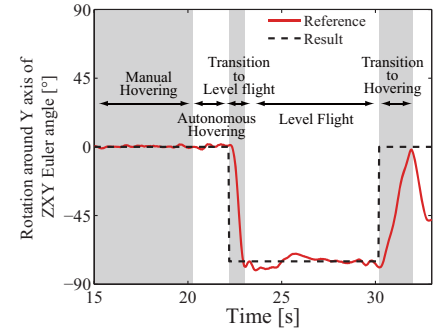


Fig. 11. Rotation around Y axis of ZXY Euler angle

greater than the aircraft weight despite of the measured altitude was not accurate enough.

Consequently, we could verify the transition and level flight with the developed quad rotor tail-sitter VTOL UAV, which does not have any control surfaces. The control of the altitude, however, was not perfectly achieved for the reason of inaccurate altitude measurement by the USS due to the grass field. This problem will be solved soon by carrying out again in the different field condition and establishing more sophisticated control strategy with a combination of several sensors so that the UAV can fly in the altitude of our scope, which is about several meters to several dozen of meters or indoors of buildings.

V. CONCLUSIONS AND FUTURE WORKS

This paper described development of two different types of quad rotor tail-sitter VTOL UAVs named here cross-type and asterisk-type of UAV. We discussed the autonomous flight control including hovering, transition and level flight of the developed quad rotor tail-sitter VTOL UAV.

Firstly, we verified attitude control capability of the two developed UAVs to analyze the effect of the slipstream which was not completely modeled in the simulation analysis in our previous research in [8]. The verification showed that the asterisk-type of UAV is superior to the cross-type of UAV in terms of the attitude control capability. Therefore, the experiment of transition and level flight was performed with the asterisk-type of UAV. As a result, the asterisk-type of UAV has succeeded in hovering, transition and level flight. In the experiment, the attitude was controlled very well and it could converge to the reference attitude, but the altitude was not perfectly controlled. Since the lift force was generated greater than the aircraft weight, the controller itself was functional. However, the measurement of the altitude seemed to have an error due to the grass field and the USS capability.

In the future, we will verify the improvement of energy efficiency of the UAV in the level flight compared with the conventional quad rotor helicopter. Moreover, we will establish more sophisticated altitude transition strategy of the quad rotor tail-sitter UAV for low altitude flight and flight indoors building. Furthermore, we will analyze aerodynamic characteristic of the developed UAV by using of wind tunnel and try to find out optimal aerodynamic configuration for hovering and level flight.

REFERENCES

- [1] B. W. McCormick, "Aerodynamics of V/STOL Flight", Academic Press, 1969.
- [2] H. Stone, G. Clarke, "Optimization of Transition Maneuvers for a Tail-Sitter Unmanned Air Vehicle (UAV)", in *Proc. 5th Australian International Aerospace Congress*, Canberra, pp. 105, 2001.
- [3] K. Kita, A. Konno, M. Uchiyama, "Transition between Level Flight and Hovering of a Tail-Sitter Vertical Takeoff and Landing Aerial Robot", *Advanced Robotics*, Vol. 24, pp. 763-781, 2010.
- [4] S. Shen, N. Michael and V. Kumar, "Autonomous Multi-Floor Indoor Navigation with a Computationally Constrained MAV", in *Proc. IEEE Int. Conf. on Robotics and Automation*, pp. 20-25, Shanghai, 2011.
- [5] M. Valenti, B. Bethke, D. Dale, A. Frank, "The MIT Indoor Multi-Vehicle Flight Testbed", in *Proc. IEEE Int. Conf. on Robotics and Automation*, Rome, 2758-2759, 2007.
- [6] L. A. Young, E. W. Aiken, J. L. Johnson, R. Demblewski, "New Concepts and Perspectives on Micro-Rotorcraft and Small Autonomous Rotary-Wing Vehicles", Ames Research Center, Moffett Field, 2002.
- [7] P. Sinha, P. Esden-Tempski, C. A. Forrette, J. K. Gibboney, G. M. Horn "Versatile, Modular, Extensible VTOL Aerial Platform with Autonomous Flight Mode Transitions", in *Proc. 2012 IEEE Aerospace Conference*, Montana, pp. 1-17, 2012.
- [8] A. Oosodo, A. Konno, T. Matsumoto, K. Go, K. Masuko, M. Uchiyama "Design and Attitude Control of a Quad Rotor Tail-Sitter Vertical Takeoff and Landing Unmanned Aerial Vehicle", *Advanced Robotics*, Vol. 26, pp. 307-326, 2012.
- [9] T. Matsumoto, K. Kita, R. Suzuki, A. Oosodo, K. Go, Y. Hoshino, A. Konno and M. Uchiyama, "A Hovering Control Strategy for a Tail-Sitter VTOL UAV that Increases Stability Against Large Disturbance", in *Proc. IEEE Int. Conf. on Robotics and Automation*, Alaska, pp. 54-59, 2010.